

TSA6213 SYSTEM ADMINISTRATION & MAINTENANCE

LAB 12

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Backing Up Data - Tar and Rsync

1. tar (i.e., tape archive) command is used to convert a group of files into an archive. Below are tar commands and their options:

```
tar option(s) archive_name file_name(s)/dirs
```

Options

c - create a new backup archive.

v - verbose mode, tar will print what it's doing to the screen.

p - preserve the permissions of the files put in the archive for restoration later.

z - compress the backup file with 'gzip' to make it smaller.

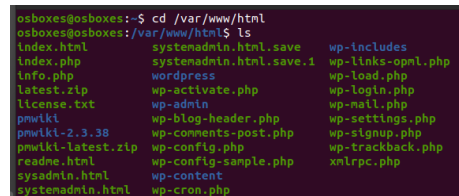
f <filename> - specifies where to store the backup

-t, --list list the contents of an archive

a) Create a backup or an archive of your web server **folder /var/www/html** using tar command. Use compression in your archive. Give the archive name as **webserver-backup.tar.gz**.

```
> cd /var/www/html
```

```
> ls
```



```
osboxes@osboxes:~$ cd /var/www/html
osboxes@osboxes:/var/www/html$ ls
index.html          systemadmin.html.save  wp-includes
index.php           systemadmin.html.save.1 wp-links-opml.php
info.php            wordpress              wp-load.php
latest.zip          wp-activate.php        wp-login.php
license.txt         wp-admin               wp-mail.php
pmwiki              wp-blog-header.php     wp-settings.php
pmwiki-2.3.38       wp-comments-post.php   wp-signup.php
pmwiki-latest.zip  wp-config.php          wp-trackback.php
readme.html        wp-config-sample.php   xmlrpc.php
sysadmin.html      wp-content
systemadmin.html   wp-cron.php
```

```
> tar -czvf webserver-backup1.tar.gz /var/www/html
```

```
osboxes@osboxes:~$ sudo tar -czvf webserver-backup1.tar.gz /var/www/html
tar: Removing leading `/' from member names
/var/www/html/
/var/www/html/wp-links-opml.php
/var/www/html/wp-includes/
/var/www/html/wp-includes/general-template.php
/var/www/html/wp-includes/certificates/
/var/www/html/wp-includes/certificates/ca-bundle.crt
/var/www/html/wp-includes/template.php
```

> ls

```
/var/www/html/xmlrpc.php
osboxes@osboxes:~$ ls
add_numbers.sh          Desktop          loop.sh          public_html      Templates
adminer-4.8.1-mysql-en.php Documents        Music           Public_html      Videos
calculate.sh            Downloads       myip1.sh        report1.txt      webserver-backup1.tar.gz
car                     filecheck.sh    myip.sh         report2.txt      webserver-backup.tar.gz
car1                    hello.sh        Pictures        Report3.txt      welcome.sh
car.sh                  lab11.txt       Public          report.txt       WindowsToLinux
```

b) View content of your archive or backup using suitable **tar** command / options

> tar -tzvf webserver-backup.tar.gz

```
osboxes@osboxes:~$ tar -tzvf webserver-backup.tar.gz
-rwxr-xr-x www-data/www-data 11753 2023-02-12 11:11 var/www/html/pmwiki/pmwiki-2.3.38/wiki
lib.d/PmWiki.TextFormattingRules
drwxr-xr-x www-data/www-data 0 2024-11-03 11:03 var/www/html/pmwiki/pmwiki-2.3.38/docs
/
-rwxr-xr-x www-data/www-data 363 2022-02-17 09:48 var/www/html/pmwiki/pmwiki-2.3.38/docs
/DOCUMENTATION.txt
-rwxr-xr-x www-data/www-data 1609 2022-02-17 09:48 var/www/html/pmwiki/pmwiki-2.3.38/docs
/INSTALL.txt
-rwxr-xr-x www-data/www-data 24 2006-02-28 20:30 var/www/html/pmwiki/pmwiki-2.3.38/docs
/.htaccess
-rwxr-xr-x www-data/www-data 18008 2012-08-17 04:26 var/www/html/pmwiki/pmwiki-2.3.38/docs
/COPYING.txt
-rwxr-xr-x www-data/www-data 1563 2022-02-17 09:48 var/www/html/pmwiki/pmwiki-2.3.38/docs
/UPGRADE.txt
```

2. Using `rsync` with suitable options, copy and backup your files/folder as below:

a) Backup your web server directory `/var/www/html` to `web-server-backup` folder in your home directory and show the statistic of the transaction. Use `ls` command to view both directory contents after completing `rsync` task.

```
> rsync -avz --stats /var/www/html/ ~/web-server-backup/
```

```
osboxes@osboxes:~$ rsync -avz --stats /var/www/html/ ~/web-server-backup/
sending incremental file list
created directory /home/osboxes/web-server-backup
./
.htaccess
index.html
index.php
info.php
latest.zip
license.txt
pmwiki-latest.zip
readme.html
sysadmin.html

Number of files: 7,650 (reg: 6,935, dir: 715)
Number of created files: 7,650 (reg: 6,935, dir: 715)
Number of deleted files: 0
Number of regular files transferred: 6,935
Total file size: 181,264,140 bytes
Total transferred file size: 181,264,140 bytes
Literal data: 181,264,140 bytes
Matched data: 0 bytes
File list size: 0
File list generation time: 0.001 seconds
File list transfer time: 0.000 seconds
Total bytes sent: 85,994,251
Total bytes received: 135,922

sent 85,994,251 bytes  received 135,922 bytes  11,484,023.07 bytes/sec
total size is 181,264,140  speedup is 2.10
```

```
> cd ~/web-server-backup/
```

```
> ls
```

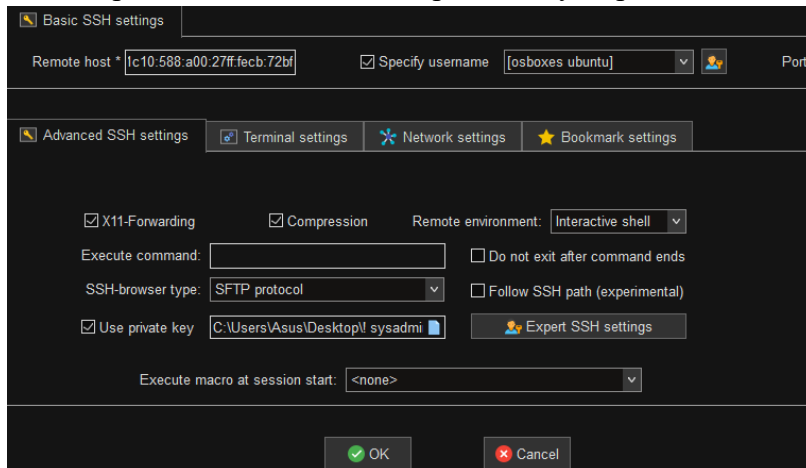
```
osboxes@osboxes:~$ cd ~/web-server-backup/
osboxes@osboxes:~/web-server-backup$ ls
index.html      readme.html      wp-blog-header.php  wp-load.php
index.php       sysadmin.html    wp-comments-post.php wp-login.php
info.php        systemadmin.html wp-config.php        wp-mail.php
latest.zip      systemadmin.html.save wp-config-sample.php wp-settings.php
license.txt     systemadmin.html.save.1 wp-content            wp-signup.php
pmwiki          wordpress         wp-cron.php          wp-trackback.php
pmwiki-2.3.38   wp-activate.php   wp-includes           xmlrpc.php
pmwiki-latest.zip wp-admin          wp-links-opml.php
osboxes@osboxes:~/web-server-backup$
```

b) Backup your web server directory `/var/www/html` to your Windows machine in Desktop/web-server-backup folder. Use MobaXTerm as `rsync` client, use compression and show statistics for `rsync`.

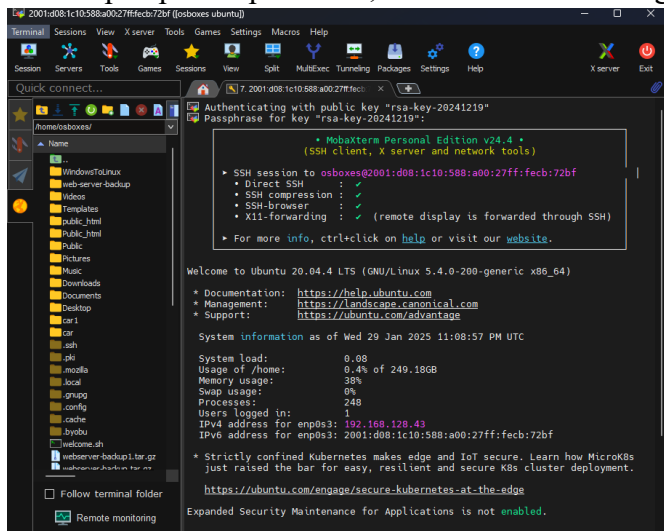
```
> ifconfig
```

```
osboxes@osboxes:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet6 fe80::a00:27ff:feeb:72bf prefixlen 64 scopeid 0x20<link>
    inet6 2001:d08:1c10:588:a00:27ff:feeb:72bf prefixlen 64 scopeid 0x0<global>
    ether 08:00:27:cb:72:bf txqueuelen 1000 (Ethernet)
    RX packets 81574 bytes 116852248 (116.8 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 5612 bytes 580269 (580.2 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

> enter ip at remote host and use private key in previous lab



> enter passphrase/password, and we are able to login

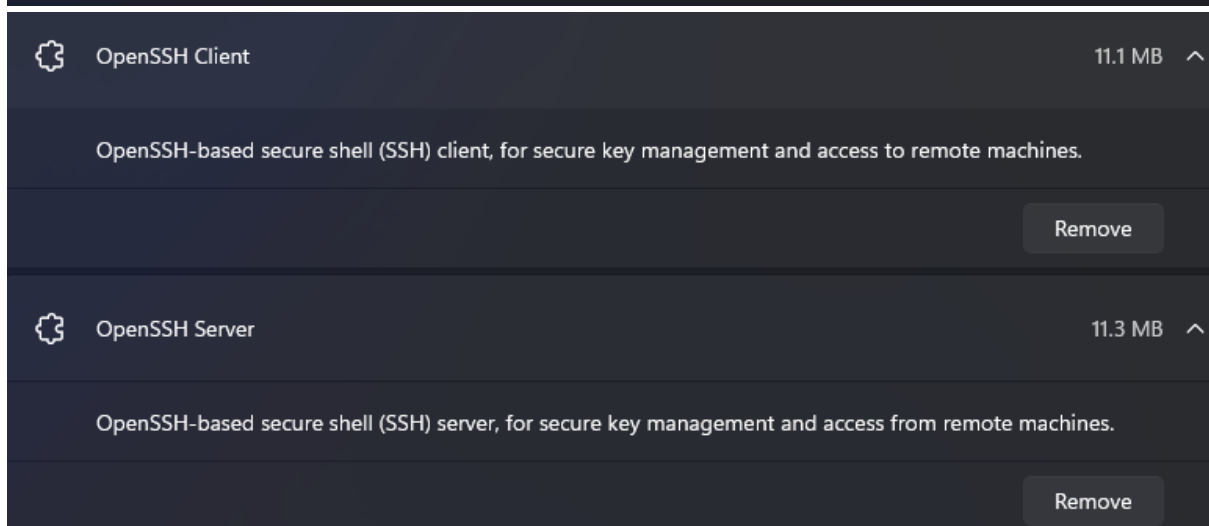
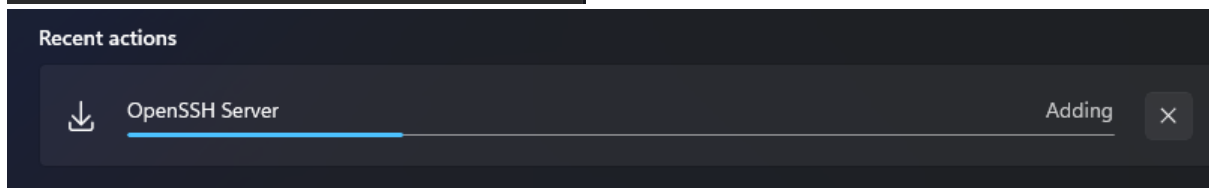
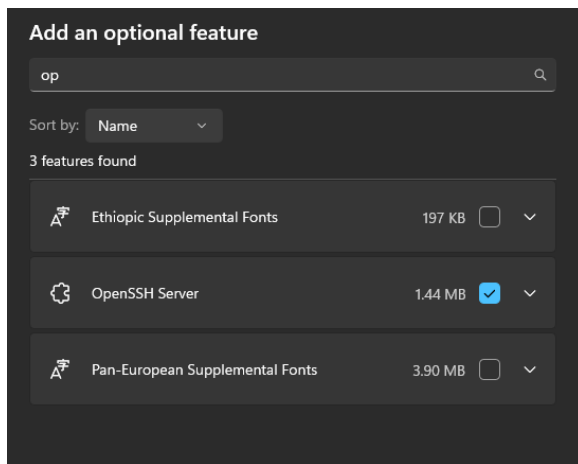


> rsync -avz --stats -e ssh /var/www/html/

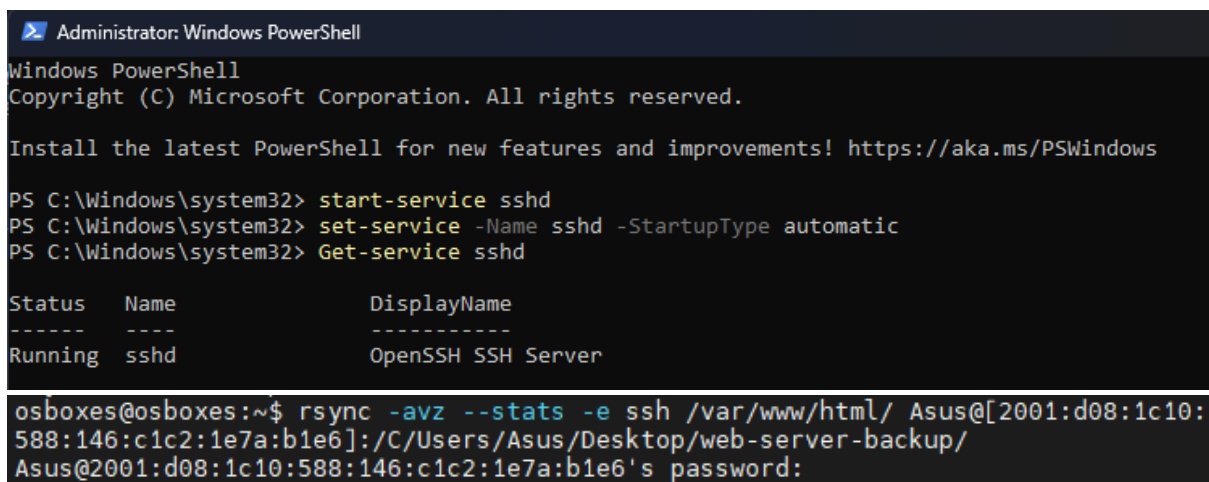
Asus@[2001:d08:1c10:588:146:c1c2:1e7a:b1e6]:/C/Users/Asus/Desktop/web-server-backup/

```
osboxes@osboxes:~$ rsync -avz --stats /var/www/html/ Users@2001:d08:1c10:588:146:c1c2:1e7a:b1e6:/Desktop/web-server-backup/
ssh: connect to host 0.0.7.209 port 22: Connection timed out
rsync: connection unexpectedly closed (0 bytes received so far) [sender]
rsync error: unexplained error (code 255) at io.c(235) [sender=3.1.3]
```

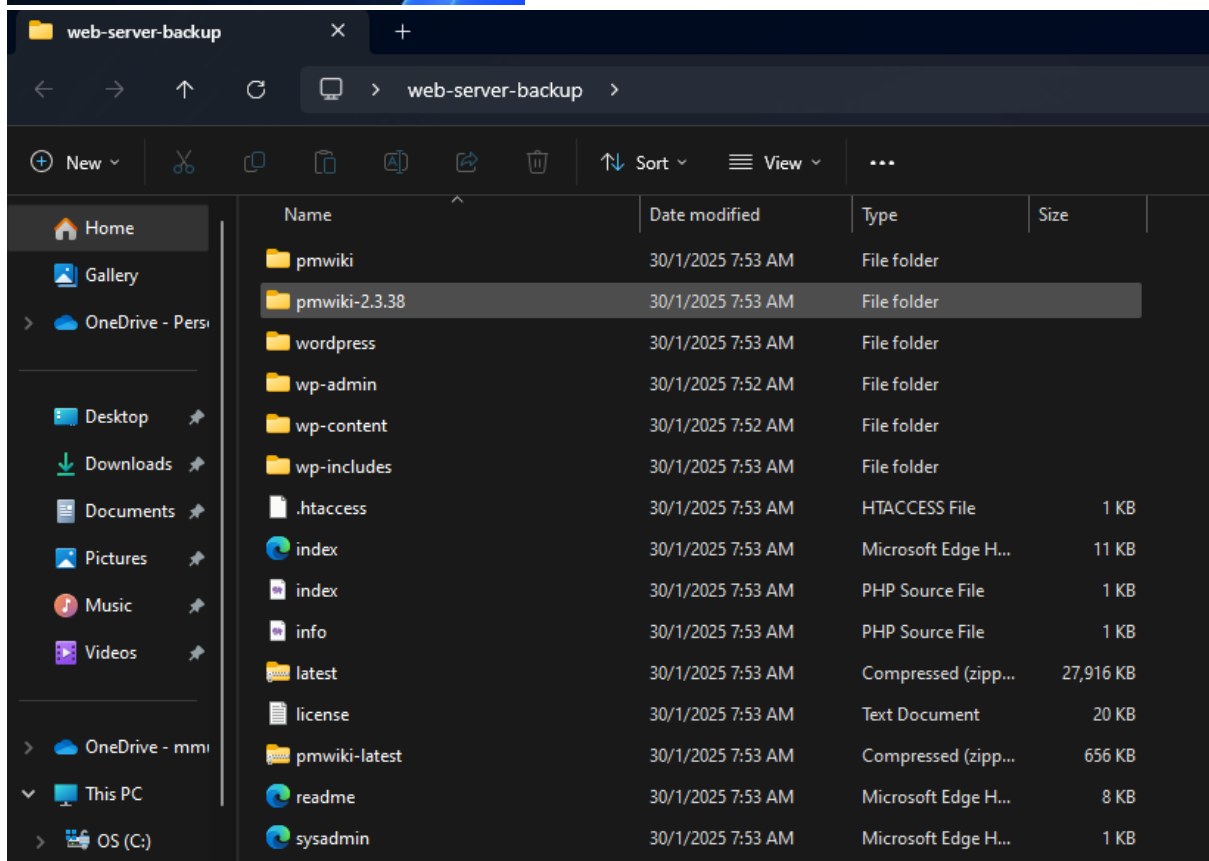
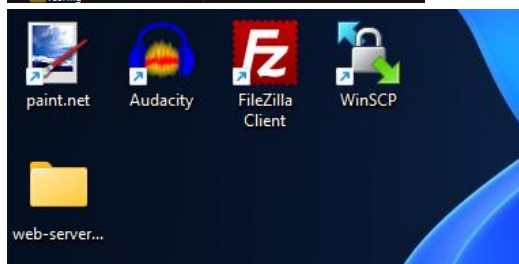
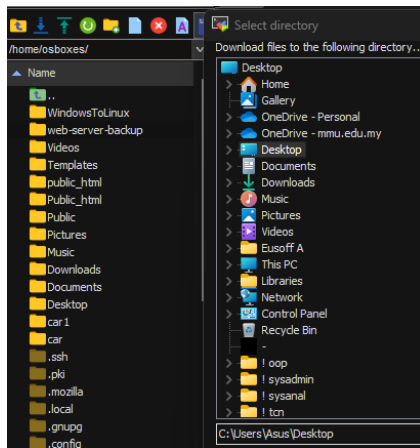
> Settings, Optional Feature, add an optional feature, OpenSSH



> run as admin



> If unable, just do it manually.



There you go.

Backup Tools and Automated Task

3. Using **rsync** and **at** command with suitable options, copy and backup your files as below:

a) Use **at** command to set the task on your **current time + 5 minutes** and in the command use **rsync** to copy your web folder `/var/www/html` to your home folder `/home/username/backup-lab12`.

> echo "rsync -avz /var/www/html/ ~/backup-lab12/" | at now + 5 minutes

```
osboxes@osboxes:~$ echo "rsync -avz /var/www/html/ ~/backup-lab12/" | at now + 5 minutes
warning: commands will be executed using /bin/sh
job 1 at Wed Jan 29 23:20:00 2025
```

```
osboxes@osboxes:~$ echo "rsync -avz /var/www/html/ ~/backup-lab12/" | at now + 5 minutes
warning: commands will be executed using /bin/sh
job 2 at Wed Jan 29 23:21:00 2025
```

b) View the **at** queue.

> atq

```
osboxes@osboxes:~$ atq
1      Wed Jan 29 23:20:00 2025 a osboxes
2      Wed Jan 29 23:21:00 2025 a osboxes
```

c) Show the final result/output using **ls** command in your home directory.

```
osboxes@osboxes:~$ ls -l ~/backup-lab12
total 0
```

> backup-lab12 is there

```
osboxes@osboxes:~$ ls
add_numbers.sh          Documents              myip.sh               report.txt
adminer-4.8.1-mysql-en.php Downloads             Pictures              Templates
backup-lab12            filecheck.sh          Public               Videos
calculate.sh           hello.sh              public_html           web-server-backup
car                     lab11.txt             Public_html           webserver-backup1.tar.gz
car1                    loop.sh               report1.txt           webserver-backup.tar.gz
car.sh                  Music                 report2.txt           welcome.sh
Desktop                 myip1.sh              Report3.txt           WindowsToLinux
```

4. Create an automated backup of your web server directory `/var/www/html` using **tar** and run at specific time using **cron** as below:

a) Create a backup script called `backup.sh` in your home directory. In the script, use **tar** command to compress your `/var/www/html` as `web-backup.tar.gz` to `/tmp` folder. Make your script executable **chmod 755 backup.sh**

> `sudo nano backup.sh`

```
GNU nano 4.8 backup.sh
tar -czvf /tmp/web-backup.tar.gz /var/www/html
```

```
osboxes@osboxes:~$ sudo nano backup.sh
osboxes@osboxes:~$ ls
add_numbers.sh          Documents
adminer-4.8.1-mysql-en.php Downloads
backup-lab12            filecheck.sh
backup.sh               hello.sh
calculate.sh            lab11.txt
```

> `sudo chmod 755 backup.sh`

```
osboxes@osboxes:~$ sudo chmod 755 backup.sh
```

b) Using **crontab -e**, write the **cron** job that executes the backup script on month of October at specific time during your lab session so that you can view the output/result.

> `crontab -e`

```
osboxes@osboxes:~$ crontab -e
no crontab for osboxes - using an empty one

Select an editor. To change later, run 'select-editor'.
 1. /bin/nano      <---- easiest
 2. /usr/bin/vim.basic
 3. /usr/bin/vim.tiny
 4. /bin/ed

Choose 1-4 [1]: 1
crontab: installing new crontab
```

> `0 14 15 10 * /home/osboxes/backup.sh`

[Run at 2:00pm on October 15]

```
GNU nano 4.8 /tmp/crontab.2YihJf/crontab
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
0 14 15 10 * /home/osboxes/backup.sh
```


c) Show the result/output using **ls -l** command in /tmp folder.

> ls -l /tmp

```
osboxes@osboxes:~$ ls -l /tmp
total 44
-rw----- 1 osboxes osboxes    0 Jan 29 22:48 config-err-R3MvBA
drwx----- 3 root    root      4096 Jan 29 22:41 snap-private-tmp
drwx----- 2 osboxes osboxes  4096 Jan 29 22:48 ssh-AIb18qxLP3G8
drwx----- 3 root    root      4096 Jan 29 22:41 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-apache2.service-geroRi
drwx----- 3 root    root      4096 Jan 29 22:42 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-colord.service-p6AeLi
drwx----- 3 root    root      4096 Jan 29 22:41 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-ModemManager.service-LM8yEf
drwx----- 3 root    root      4096 Jan 29 22:41 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-switcheroo-control.service-5Hk81i
drwx----- 3 root    root      4096 Jan 29 22:41 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-systemd-logind.service-ltsmkh
drwx----- 3 root    root      4096 Jan 29 22:41 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-systemd-resolved.service-e73fVh
drwx----- 3 root    root      4096 Jan 29 22:41 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-systemd-timesyncd.service-Gy99Ph
drwx----- 3 root    root      4096 Jan 29 22:42 systemd-private-74a48648527a4fd8a939e3c86145f
0ab-upower.service-5kG2gj
drwx----- 2 osboxes osboxes  4096 Jan 29 23:24 tracker-extract-files.1000
```